# Henry Klatt

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## Education

**Ph.D. in Mathematics**, The George Washington University 2022–present
Expected May 2027. Advisor: Valentina Harizanov.
Dissertation working title: *Effective Nonstandard Algebra*.
Specialty examination: Passed with distinction, "Forcing in Computability Theory" (Mar. 22, 2024).

**M.A. in Mathematics**, The George Washington University 2024

**B.S. in Mathematics**, Iowa State University 2022
*Magna cum laude*.
Undergraduate thesis: *Language, Models, Ultraproducts, and the Hyperreals* (2021).
Advisor: Tim McNicholl.

## Research Appointments and Fellowships

**Research Fellowship**, Hausdorff Research Institute for Mathematics Fall 2025
Trimester programme: "Definability, Decidability, and Computability," Bonn, Germany.

## Research Interests

Mathematical logic, primarily computability theory, with a focus on ultraproduct-like constructions in model theory and computability theory. Additional interests include knot theory, algorithmic learning theory, and number theory.

## Preprints and Work in Preparation

### Preprint

R. Dimitrov, V. Harizanov, **H. Klatt**, and K. Srinivasan. *On Cohesive Products of Fields* (2026).
Submitted to the *Journal of Symbolic Logic*. arXiv:2604.09965.

### Work in Preparation

V. Harizanov and **H. Klatt**. *HKSS Completeness of Quandles, Racks, and Keis*.
In preparation.

V. Harizanov, M. "Turbo" Ho, **H. Klatt**, M. Kukla, and K. Srinivasan. *Cohesive Powers of Abelian Groups*.
In preparation.

V. Harizanov, **H. Klatt**, and K. Srinivasan. *Generalized Cohesive Products*.
In preparation.

W. Calvert, R. Dimitrov, V. Harizanov, **H. Klatt**, and K. Srinivasan. *Automorphisms of Effective Vector Spaces*.
In preparation.

## Teaching Experience

**Instructor of Record**, The George Washington University Summers 2023, 2025, 2026
Math and Politics. Independently responsible for course delivery, assessment, and grading.

**Graduate Teaching Assistant**, The George Washington University 2022–present
Calculus I, II, and III; Math and Politics; Mathematical Ideas.

## Talks

### Invited Conference Talks

*Cohesive Products of Galois Extensions* — Sept. 7–11, 2026
Computability, Complexity, and Randomness 2026, University of Leeds.

*Countable Ultraproducts* — May 29, 2026
Logicón, Universidad Nacional Autónoma de México.

*Cohesive Products of Fields* — Jan. 5, 2026
Joint Mathematics Meetings, ASL Special Session on Computability and Its Applications, Washington, DC.

*Cohesive Powers of Number Fields* — Apr. 6, 2025
AMS Eastern Sectional Meeting, hosted by the University of Connecticut, Hartford.

*Quandles are Complicated* — Aug. 1, 2024
FRG Workshop on Definability, Decidability, and Computability, Harvard University.

*A maximally complicated isomorphism problem* — May 4–5, 2024
AMS Western Sectional Meeting, San Francisco State University.

*Quandles are Complicated* — Dec. 10, 2023
Knots in Washington, The George Washington University.

### Seminar Talks

*Cohesive Products of Galois Extensions* — Sept. 22, 2026
Logic Seminar, Iowa State University.

*Cohesive Powers of Galois Extensions* — Apr. 23, 2026
Online Logic Seminar.

*More on cohesive products of fields, Part II* — Feb. 11, 2026
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*More on cohesive products of fields* — Feb. 4, 2026
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*Introduction to Cohesive Powers* — Nov. 14, 2025
Junior Seminar, Hausdorff Research Institute for Mathematics.

*Types, existential closures, and cohesive powers in the study of arithmetic* — Sept. 2025
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*Why doesn't Chevalley's theorem solve the halting problem?* — Jan. 24, 2025
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*Smaller Ultraproducts, part 2* — Oct. 16, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*Smaller Ultraproducts: A Survey* — Oct. 2, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*n-generic and more: how far can forcing take us?* — Mar. 27, 2024
University Seminar on Computability, Complexity, and Algebraic Structure, GW.

*The isomorphism problem for quandles is complicated* — Sept. 6, 2023
Logic Seminar, The George Washington University.

*Infinitesimal Calculus* — Oct. 3, 2022
Louise Hay Logic Seminar, University of Illinois at Chicago.

*Ultrapower construction of the hyperreals* — Sept. 15, 2022
Logic Seminar, The George Washington University.

*Some fruit from set-theoretic pluralism* — Apr. 6, 2022
Logic Seminar, Iowa State University.

*How Effective is Forcing?* — Nov. 30, 2021
Logic Seminar, Iowa State University.

## Professional Service

### Conference and Programme Organization

**Computability of Algebraic Structures** (scheduled) — June 20–25, 2027
Lead organizer; BIRS Workshop, Casa Matemática Oaxaca (CMO).

**Special Session on Mathematical Logic** — Oct. 3–4, 2026
Co-organizer; AMS Eastern Sectional Meeting, Washington, DC.

**Special Session: DDC@DC** — Jan. 4–7, 2026
Co-organizer; Joint Mathematics Meetings, Washington, DC.

**Seminar on Computable Structure Theory** — Fall 2025
Organizer; trimester programme: "Definability, Decidability, and Computability."
Hausdorff Research Institute for Mathematics.

**Special Session on Computability, Complexity, and Algebraic Structure** — Apr. 6–7, 2024
Co-organizer; AMS Spring Eastern Sectional Meeting, Howard University, Washington, DC.

### Seminar Organization

**Logic Seminar**, The George Washington University — 2022–present
Co-organizer.

### Reading Group Organization

**Computable Structure Theory** — Sept.–Dec. 2025
Co-organizer; Hausdorff Research Institute for Mathematics.

**Commutative Algebra** — May–July 2025
Co-organizer; The George Washington University.

## Honors and Research Support

**Travel Grants**, Association for Symbolic Logic — 2024, 2025, 2026

**Summer Research Support** — 2023, 2024
Supported through advisor Valentina Harizanov's National Science Foundation grant.